Cross-Regional Generalization of U.S.-Trained Geospatial Imagery Models for Energy Infrastructure Detection in Russia and China

Zoe Gates
Department of Data Science and Environmental Science
University of Virginia

Course: Deep Learning in Environmental Science
Instructor: Antonios Mamalakis
April 2026

ABSTRACT

This study investigates the ability of convolutional neural networks (CNNs) to generalize internationally across satellite imagery. A custom CNN and a pretrained ResNet-50 model were both trained on U.S. power plant data to classify four energy plant types: Natural gas, hydro, solar, and wind. To evaluate cross-regional performance, a test dataset was constructed using Sentinel-2 imagery from Russia and China, extracted programmatically via geographic coordinates. The ResNet-50 model achieved the strongest validation performance (~75% accuracy) on U.S. data, but its performance dropped significantly on the external test set (~57% accuracy). The results suggest that geographic variation, infrastructure gaps, sensor differences, and resolution discrepancies all contribute to reduced performance. While this work shows potential for transfer learning in geospatial intelligence (GEOINT) applications, it emphasizes the importance of consistency in dataset design. Furthermore, it reveals the tradeoff between spatial detail and contextual information in geospatial classification tasks.

INTRODUCTION

The state of international relations is dynamic, and with current and recent wars in Iran, Ukraine, and Palestine, energy has remained at the center of attention [1]. The balance and placement of energy sources and technologies globally is a major security concern among powers such as the United States, Russia, and China. Their ability to power their nations and militaries significantly impacts their interactions and leverage in times of conflict. In this context, access to reliable information about foreign infrastructure plays a critical role in reducing uncertainty and informing strategic decision-making. From a realist perspective, states work to improve their situational awareness in order to effectively pursue their national security interests.

Advances in satellite imagery and machine learning have created new opportunities to automate and improve this process, enabling large-scale analysis of global energy infrastructure. The National Geospatial-Intelligence Agency recognizes the importance of integrating AI into its workflows to support speed and scale of data processing and analysis (GEOINT AI) [2]. However, these efforts and supervised learning methods depend heavily on labeled data. Labeling satellite imagery at scale is expensive, slow, and inconsistent [3]. So, discovering a new, automated way to process and evaluate new, unseen geographic data is vital to supporting GEO INT efforts. Furthermore, being able to monitor energy distributions of adversaries is helpful for achieving a complete picture of international adversary economic and military potential. The question that arises is: Can a model trained on U.S. satellite imagery reliably identify energy infrastructure in other regions of interest, such as Russia and China? To promote reproducibility, our code and data processing pipeline are publicly available at https://github.com/gatesz33/Global_Plants.

DATA AND METHODS

Training Data

The data used for training was Power Plant Satellite Imagery Dataset (US Power Plants NAIP/LANDSAT8) released in 2017 [4]. The data is commonly utilized in the energy industry for instance segmentation, semantic segmentation, object detection, and classification tasks. It contains 4,454 instances at two different resolutions (1m/pixel and 30m/pixel). The dataset contains multiple instances per plant, typically including both high-resolution National Agriculture Imagery Program (NAIP) imagery and medium-resolution Landsat-8 (American Earth observation satellite) imagery. In all, the dataset contains 8908 images with 5185 unique power plants. While all objects belong to a single ‘power plant’ class for detection, each instance is associated with one of 12 fuel types [5].

Training Data Preprocessing

The images were in GeoTIFF format, often used for Remote Sensing Imagery (RSI). GeoTIFF images often vary in spatial dimensions and may contain multiple spectral bands beyond the baseline RGB [6]. This data requires additional preprocessing such as resizing and band selection to ensure consistent input for convolutional neural network training. Images from the training dataset were converted from GeoTIFF format to RGB JPEG images and resized to 224×224 pixels to match the input requirements for the models.

The images were not explicitly labeled by fuel type; however, this information was embedded in the file names. As a result, we directly extracted fuel type labels from the file names. Due to observed class imbalance across fuel types, we restricted the classification task to the four categories with the greatest number of image samples: Natural gas, solar, hydro, and wind.

We created image generators that normalized the data pixel-by-pixel and applied random transformations during training. These random augmentations increased the diversity of data and helped reduce overfitting, allowing the models to generalize more effectively. This was key for eventual cross-regional generalization and anticipated geographic domain shifts.

Test Data Construction

We constructed the test Russia and China power plant dataset using the World Resources Institute Global Power Plant Database and Sentinel-2 (European Earth observation satellite) imagery. The WRI Global Power Plant database tracks approximately 30,000 power plants from 164 countries and includes 11 categories of fuel type [7]. Each power plant is associated with coordinates and information on plant capacity, generation, ownership, and fuel type. We accessed Sentinel-2 imagery using Google Earth Engine (GEE), which allows the user to pull specific areas over customizable date ranges. Custom image patches were extracted based on specific geographic coordinates from the WRI dataset. We used a buffer measure of 500m to match the spatial context that the CNN was training on (approximately 1 km × 1 km). We were able to generate 1089 labeled images (China = 844, Russia = 245).

To maintain consistency across U.S. training and global testing, we kept only natural gas, hydro, solar, and wind plants in the test set.

Model Construction, Training, and Selection

We built two Convolutional Neural Network (CNN) models and trained them on the U.S. power plant dataset. This was done to compare a model built from scratch with a pre-trained CNN and select the higher-performing model for further regional generalization.

The initial model, named ‘model’, consisted of 13 layers, including four convolutional-pooling blocks followed by fully connected layers. It utilized the ReLU and softmax activation functions to introduce nonlinearity and assign probabilities to logits during training. The model was trained using the Adam optimizer with categorical cross-entropy loss. Early stopping with a patience of 3 epochs was implemented to reduce overfitting.

The second model, ‘model2’, utilized a ResNet-50 architecture pretrained on the ImageNet dataset. This pretraining allows the model to learn low and mid-level features such as edges, textures, shapes, and patterns from approximately 14 million images across 1000 classes, improving overall performance and training efficiency. The ResNet-50 architecture incorporates residual (skip) connections to preserve important information and mitigate the vanishing gradient problem [8]. The model was trained with a frozen backbone to retain pretrained feature representations while reducing overfitting and training time. It learned using the Adam optimizer with categorical cross-entropy loss and early stopping with a patience of 3 epochs.

Testing and Interpreting

The CNN model was evaluated on unseen test data using accuracy, precision, recall, and F1-score to assess performance. While accuracy provides an overall measure of correctness, precision, recall, and F1-score offer class-specific insights into prediction quality and detection. This was helpful because even after restricting our classification to four classes, we had slightly uneven distributions across classes.

After generating predictions, an interactive geospatial visualization was created using the Folium library. Power plant locations across Russia and China were plotted on a Leaflet map and color-coded based on the model’s predicted class. An interactive feature was included to display the true vs. predicted label for each location upon user interaction.

RESULTS

Phase 1: U.S. Landsat 8 Training of Basic CNN vs. ResNet-50 Architectures

After training both models and monitoring validation accuracy, the ResNet-50 model (‘model2’) achieved a higher validation accuracy (~76%) compared to the baseline CNN (‘model’) (~70%). Because of this significant difference in accuracy, the ResNet-50 architecture was selected for phase 2.

Basic CNN

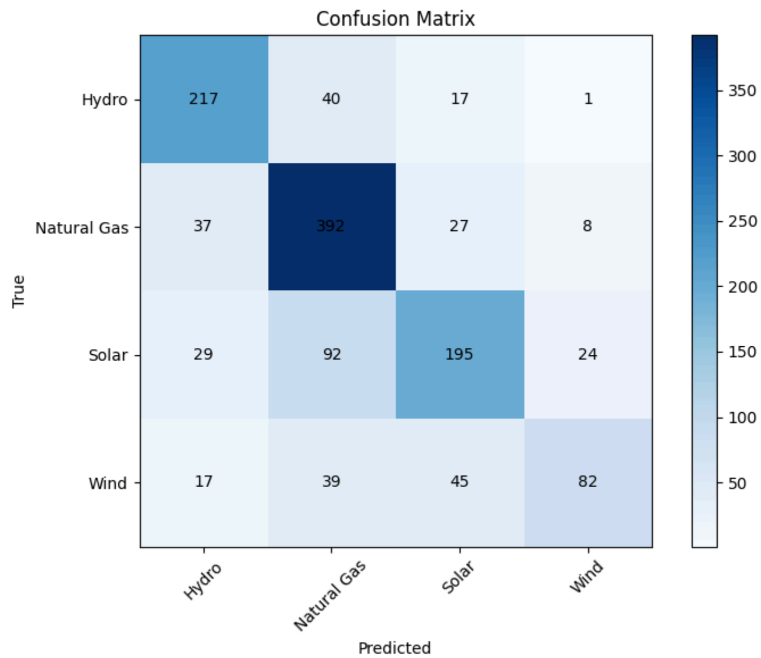

Fig. 1. Basic CNN model (‘model’) confusion matrix for unseen U.S. data.

	precision	recall	f1-score	support
Hydro	0.72	0.79	0.75	275
Natural Gas	0.70	0.84	0.76	464
Solar	0.69	0.57	0.62	340
Wind	0.71	0.45	0.55	183
accuracy			0.70	1262
macro avg	0.70	0.66	0.67	1262
weighted avg	0.70	0.70	0.69	1262

Fig. 2. Precision, recall, and f1-score results for ‘model’ on unseen U.S. data.

Natural gas is the category that was best classified by ‘model’ (F1 = 0.76). This was due to a high recall score, so more natural gas plants were successfully found. Hydro plants had the most true positive predictions. The most challenging class for this model was wind because of low recall. This means most of the wind images were missed (~55%). This is likely because wind power infrastructure is less distinctive and centralized than the other plants and facilities. Most wind plants were misclassified as solar facilities.

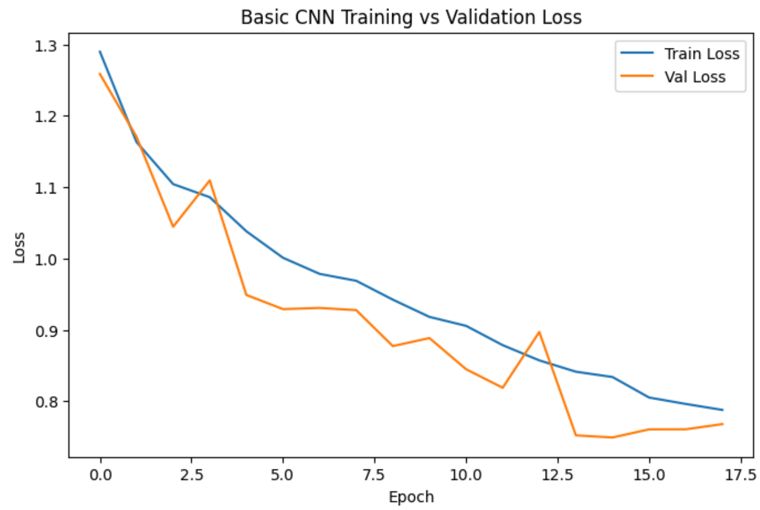

Fig. 3.Basic CNN model (model) training and validation loss over epochs

Based on Fig 3. The basic CNN model, 'model', trained healthily. Training and validation loss converged by the end and both steadily decreased over epochs. The close alignments between these metrics suggests no overfitting and good generalization ability.

Pretrained ResNet-50

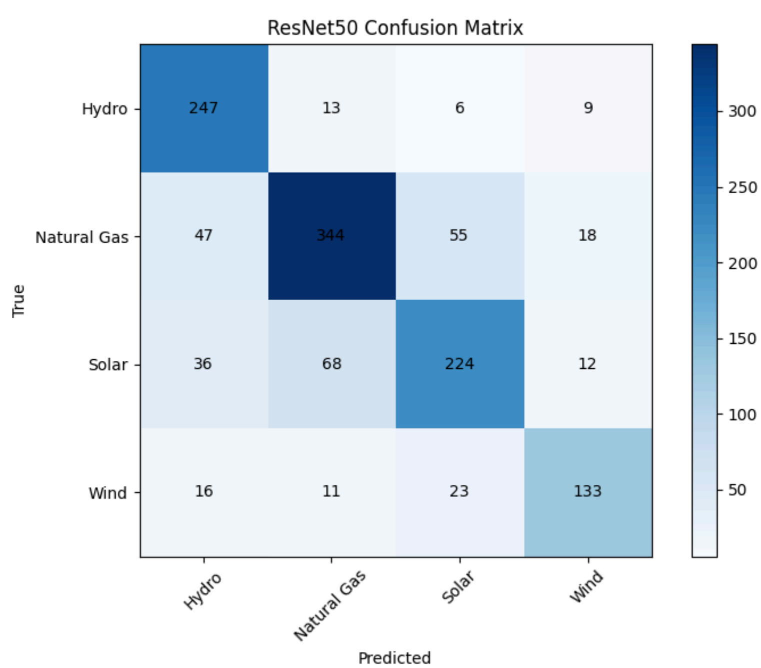

Fig. 4. ResNet-50 model (‘model2’) confusion matrix for unseen U.S. data.

	precision	recall	f1-score	support
Hydro	0.71	0.90	0.80	275
Natural Gas	0.79	0.74	0.76	464
Solar	0.73	0.66	0.69	340
Wind	0.77	0.73	0.75	183
accuracy			0.75	1262
macro avg	0.75	0.76	0.75	1262
weighted avg	0.75	0.75	0.75	1262

Fig. 5. Precision, recall, and f1-score results for ‘model2’ on unseen U.S. data.

For ‘model2’ hydro demonstrated the strongest performance (F1 = 0.80), while solar was the most challenging class (F1 = 0.69). This indicates that the model often failed to correctly predict and identify solar instances. We see evident confusion between solar and natural gas in Fig. 4. This indicates that solar features are less distinct and more easily confused with large-scale industrial structures present in natural gas facilities. It’s possible that the resolution isn’t high enough, making both solar panels and clusters of natural gas buildings appear similar in coarse pattern recognition.

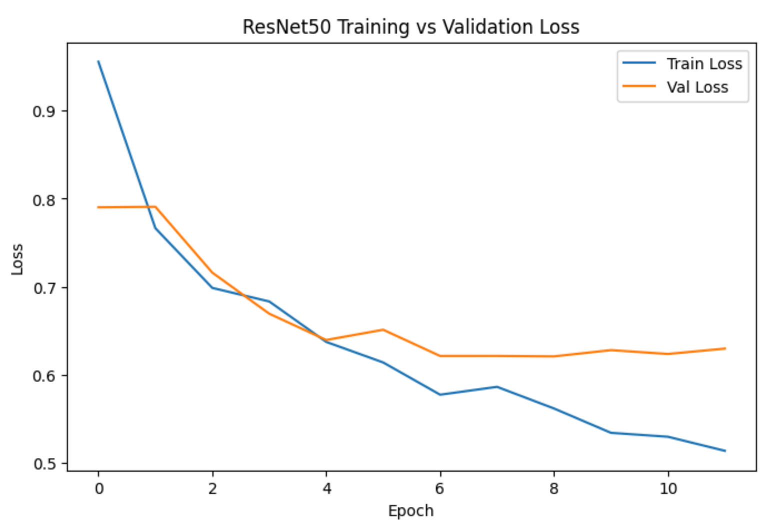

Fig. 6. ResNet-50 model (model2) training and validation loss over epochs.

Despite minor inconsistencies, model2 trained effectively, as indicated by the decreasing and stabilization of both training and validation loss. It achieved a prediction accuracy significantly above random (25%). It's likely that the residual connections and ImageNet pretraining allowed for important information to be preserved and more easily learned in training.

Phase 2: Cross-Regional Generalization and Evaluation

Testing 'model2' on our Russia/China yielded 56.93% accuracy. This is significantly lower than the 75% validation accuracy achieved by 'model2'.

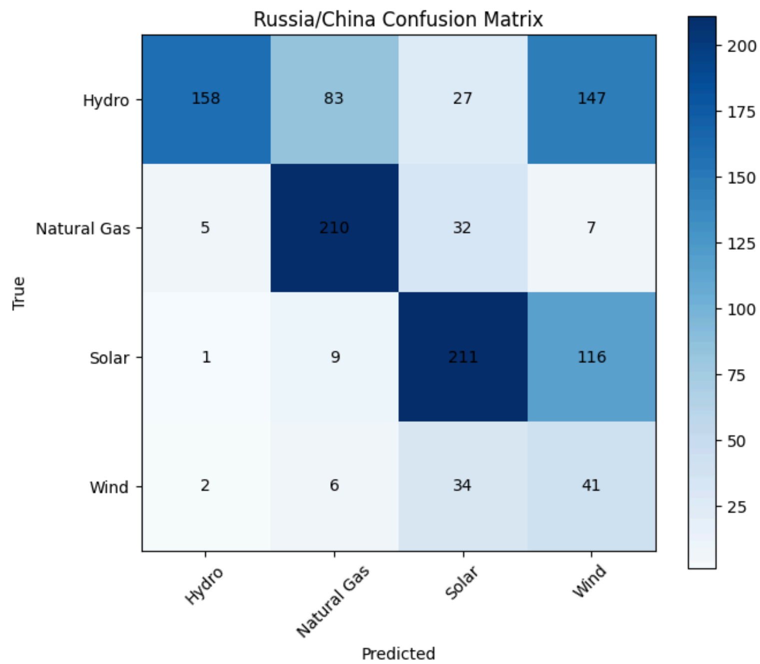

Fig. 7. Correlation matrix of 'model2' predictions vs. truths for plant type.

	precision	recall	f1-score	support
Hydro	0.95	0.38	0.54	415
Natural Gas	0.68	0.83	0.75	254
Solar	0.69	0.63	0.66	337
Wind	0.13	0.49	0.21	83
accuracy			0.57	1089
macro avg	0.61	0.58	0.54	1089
weighted avg	0.75	0.57	0.60	1089

Fig. 8. Precision, recall, and f1-score results for 'model2' on Russia and China plant data.

Based on these results, we can see that the best performing category was natural gas plant classification. These are large industrial structures that are very distinctive in design. Nearly 83% of natural gas facilities were discovered by the model. The lowest performing fuel type was wind with a precision of 0.13. This may indicate that wind farm visual characteristics vary significantly across regions, leading to poor generalization and increased false positives (only 13% of wind predictions were correct). Additionally, only 38% of hydro plants were correctly found. According to Fig. 7, many were misclassified as wind plants. This shows that the wind class in general struggled, most likely due to the delicate appearance of windmills and reliance on background context. Additionally, since wind farms can be on land or on sea, it would make sense that they were most confused with solar and hydro plants.

DISCUSSION

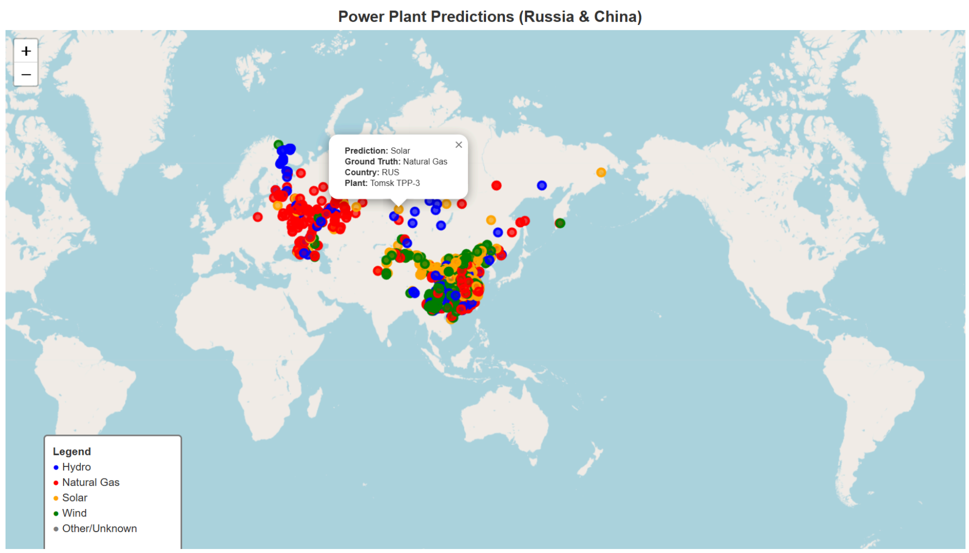

Fig. 9. Interactive Leaflet map of energy plants in Russia and China, colored by fuel type .

The generalization was limited under four domain shifts: Geographic, sensor source, resolution, and dataset. The intentional geographic shift was most likely the strongest source of inconsistencies. The world varies greatly, and the way plants look in the U.S. doesn't necessarily align with the way they look across the globe. Differences in placement, architecture, surrounding terrain, and cloud coverage were most likely also drivers of confusion. However, we definitely see some consistency across predictions. There is plenty of space for improvement despite inevitable differences.

The shift from Landsat-8/NAIP imagery to Sentinel-2 imagery posed an obstacle for the model in the prediction phase. These sensors differ in spectral characteristics and image quality, which can impact how features are represented and learned. The resolutions we trained on were also not exactly the resolutions we tested. This is because we had to either use mid-resolution Landsat-8 data without NAIP vs. higher resolution Sentinel-2 for testing. This was a decision with tradeoffs we had to make. The shift most likely added harmful noise to the predictions; however, using Landsat-8 without NAIP to test would have also served as a sensor domain shift. We deemed preservation of edges and textures to be a priority. Based on the results, fine spatial detail was key for correctly predicting wind and solar facilities. These categories broke under cross-regional generalization pressure.

Lastly, the training dataset was a carefully curated, peer-reviewed, and well-respected dataset that has been around for a significant amount of time. We constructed the test data ourselves; therefore, it's possible that there were inconsistencies with patches and the images pulled from Sentinel-2. The quality of our data highly depends on the accuracy of the coordinates in the WRI database and the success of our image pulls.

CONCLUSIONS

Overall, we saw that a CNN could be successfully trained on U.S. energy infrastructure and then partially generalized to hold-out international image data. Despite the nearly 18% drop in performance, the model still performed 32% above random baseline, indicating that some transferable features of energy infrastructure were successfully learned.

However, this level of performance remains insufficient for reliable geospatial intelligence applications. The results emphasize key limitations related to domain shifts, including differences in geography, sensor characteristics, resolution, and data construction.

Future Direction

Future improvements could include increasing training data for underrepresented classes such as wind, as well as incorporating more consistent and higher-resolution imagery for Russia and China to improve consistency in fuel-type evalution. Additionally, fine-tuning a ResNet50 model with a partially unfrozen backbone could allow pretrained weights to better adapt to satellite imagery during training.

Additionally, we were only able to evaluate a limited set of patch sizes due to computational and data-loading constraints in Google Colab. However, significant changes in test accuracy when toggling the buffer between 400m and 500m suggests that spatial context may play an important role in fuel-type classification. Future work should evaluate multiple buffer sizes across training and testing (maintaining consistency) to identify the optimal patch scale for distinguishing power plant types cross-regionally.

ACKNOWLEDGEMENTS

I would like to thank Professor Antonios Mamalakis for guidance and mentorship throughout the semester and this project, and Hiep Dang for support and feedback as a teaching assistant. I also acknowledge the use of the Global Power Plant Database provided by the World Resources Institute and the Power Plant Satellite Imagery Dataset.

REFERENCES

[1] The Guardian, "More than oil prices: Iran war threatens to reshape global energy order," 2026.
Available:
https://www.theguardian.com/business/2026/apr/26/more-than-oil-prices-iran-war-threatens-to-reshape-global-energy-order
[2] National Geospatial-Intelligence Agency, "GEOINT Artificial Intelligence,"
Available: https://www.nga.mil/news/GEOINT_Artificial_Intelligence_.html
[3] S. Chan, W. Zhou, Y. Lei, et al., “Sparse point annotations for remote sensing image segmentation,” Scientific Reports, vol. 15, p. 27347, Jul. 2025. doi: 10.1038/s41598-025-12969-6.
[4] K. Bradbury et al., "Power Plant Satellite Imagery Dataset," Aug. 2017. [Online]. Available: https://figshare.com/articles/dataset/Power_Plant_Satellite_Imagery_Dataset/5307364. doi: 10.6084/m9.figshare.5307364.v1.
[5] Dataset Ninja, "Power Plant Satellite Imagery Dataset," Dataset Ninja, 2026. [Online]. Available: https://datasetninja.com/power-plant-satellite-imagery-dataset. [Accessed: Apr. 26, 2026].
[6] NASA Earth Science Data and Information System (ESDIS) Standards Coordination Office, “GeoTIFF,” GeoTIFF page
, Sep. 14, 2019. Accessed: Apr. 27, 2026.
[7] Global Energy Observatory, Google, KTH Royal Institute of Technology in Stockholm, Enipedia, and World Resources Institute, “Global Power Plant Database,” 2018. [Online]. Available: https://datasets.wri.org/datasets/global-power-plant-database. [Accessed: Apr. 26, 2026].
[8] Ultralytics, “What is ResNet-50 and what is its relevance in computer vision?” 2025. [Online]. Available:
https://www.ultralytics.com/blog/what-is-resnet-50-and-what-is-its-relevance-in-computer-vision. [Accessed: Apr. 26, 2026].